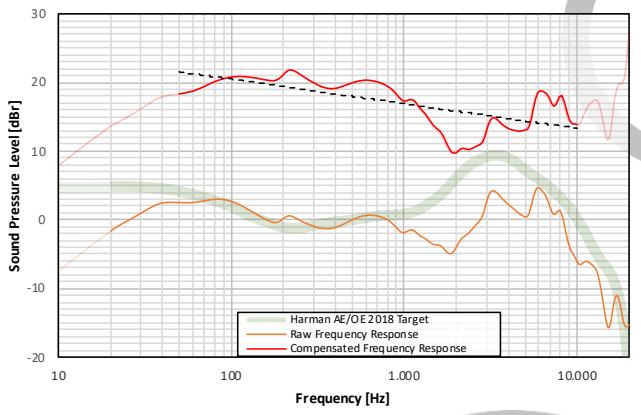
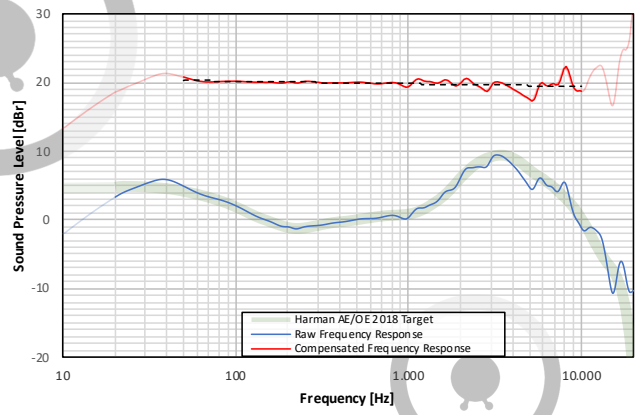
ORATORY1990

EQ setting for

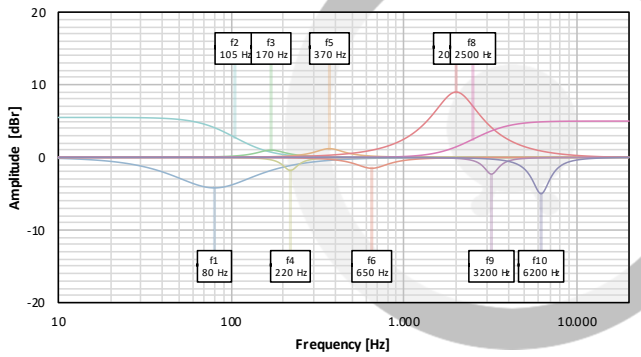
SPL Frequency Response without EQ



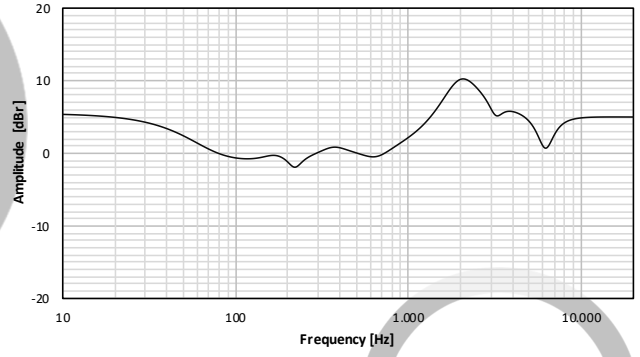
SPL Frequency Response with EQ



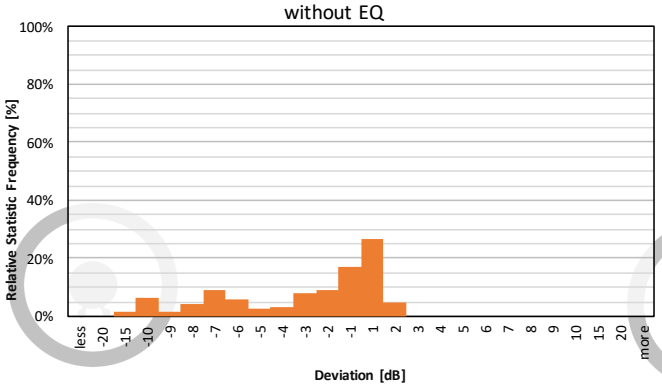
EQ Curve Individual Filters



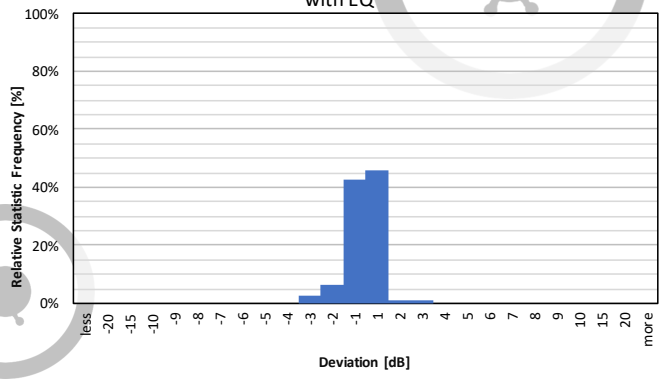
EQ Curve total



Error Curve Histogram without EQ



Error Curve Histogram with EQ



Filter Settings	Filter Type	Frequency	Gain	Q-Factor	BW
Band 1	PEAK	80 Hz	-4,2 dB	0,7	1,92
Band 2	LOW_SHELF	105 Hz	5,5 dB	0,71	1,89
Band 3	PEAK	170 Hz	1,0 dB	2,0	0,71
Band 4	PEAK	220 Hz	-1,8 dB	4,0	0,36
Band 5	PEAK	370 Hz	1,2 dB	2,0	0,71
Band 6	PEAK	650 Hz	-1,5 dB	1,8	0,79
Band 7	PEAK	2000 Hz	9,0 dB	1,1	1,27
Band 8	HIGH_SHELF	2500 Hz	5,0 dB	0,71	1,89
Band 9	PEAK	3200 Hz	-2,3 dB	4,0	0,36
Band 10	PEAK	6200 Hz	-5,0 dB	3,0	0,48

Preamp gain:
-10,3 dB

Deviation from Target	
Before EQ	After EQ
2,94 dB	0,40 dB

Preference Rating*	
Before EQ	After EQ
47/100	100/100

*preference rating prediction based on:

[1] S. Olive et al: "A Statistical Model That Predicts Listeners' Preference Ratings of In-Ear Headphones: Part 1" (2017)
[2] S. Olive et al: "A Statistical Model That Predicts Listeners' Preference Ratings of In-Ear Headphones: Part 2" (2017)
[3] S. Olive et al: "A Statistical Model That Predicts Listeners' Preference Ratings of Around-Ear and On-Ear Headphones" (2018)

The normalized preference ratings are used, where zero deviation from target equals a preference rating of 100

measured on GRAS45BC-10 KEMAR
GRAS KB5000/5001 anthropometric pinnae
APx515 or APx526 Signal Generator/Analyzer
APx1701 amplifier

v2.9
by oratory1990
08.04.24